1966	First spacecraft to orbit another world (*Luna 10* to the Moon)
1966	First successful soft landing on another world (*Luna 9* to the Moon)
1970	First robot mission to return α sample from another world (*Luna 16* to the Moon)
1970	First roving vehicle on another world (*Luna 17* to the Moon)
1971	First soft landing on another planet (*Mars 3* to Mars)
1972	First scientifically successful landing on another planet (*Venera 8* to Venus)
1980-1981	First approximately year-long manned spaceflight (*comparable to Mars flight time*) (*Soyuz 35*)
1983	First full orbital radar mapping of another planet (*Venera 15 to Venus*)
1985	First balloon station deployed in the atmosphere of another planet (*Vega 1 to Venus*)
1986	First close cometary encounter (*Vega 1 to Halley's Comet*)
1986	First space station inhabited by rotating crews (*Mir*)

UNITED STATES

1958	First scientific discovery in space —Van Allen radiation belt (*Explorer 1*)
1959	First television images of the Earth from space (*Explorer 6*)